Lab 4 VibeCoding - LLM Session Log

Project: SETAPESU26/48_whack-a-mole | SRN: PES1UG23AM287 | LLM: Claude (claude.ai)

Prompt 1 (initial)

Student: Attached the Lab 4 handout PDF and the repo link https://github.com/SETAPESU26/48_whack-a-mole. Asked: explain each step and complete this assignment with proper deliverables.

Claude: Read the repo README (4 tasks: collision bug, game-over screen, replay with difficulty, sound) and cloned the repo to inspect main.py, game/game_engine.py and game/hole.py. Found the root cause of Task 1: Hole.hit_size=150 gives square hit-boxes larger than the 125/120 px hole spacing, and _handle_click iterated over every hole without stopping. Reproduced it: one click on the border between two active moles returned score 2 on the original code.

Task 1 - Collision detection

Student: Fix so one click can score at most one point.

Claude: Replaced square hit-box with a circular hit area (radius 40, same as the drawn hole). _handle_click now collects holes containing the click, picks the closest one, whacks only it, and returns. Added tests/test_collision.py, which sweeps the whole board and asserts a click never scores more than 1. Passed. Commit: 'Task 1: Fix double-score collision'.

Task 2 - Game over screen

Student: Show the final score when the timer reaches zero and wait for input instead of printing to console.

Claude: Added a dim overlay with 'Time's Up!', final score and misses; removed the console print. handle_event now processes input in the game-over state. main.py exits when engine.quit_requested is set. Rendered a screenshot headless to confirm layout. Commit: 'Task 2: Add game-over screen'.

Task 3 - Replay + difficulty

Student: After game over let the player choose Easy / Medium / Hard or exit.

Claude: Added DIFFICULTIES table, start_round(difficulty) that fully resets score, misses, timer and holes, and four buttons (Easy, Medium, Hard, Exit) usable by mouse or keys 1/2/3/Esc. Added tests/test_replay.py covering keyboard, mouse, ignored keys, and difficulty ordering. Passed. Commit: 'Task 3: Add replay with difficulty selection'.

Task 4 - Sound

Student: Add sounds for successful whack, miss and round end.

Claude: New game/sounds.py synthesises three tones in code (no asset files) and degrades gracefully if no audio device exists. Engine plays them on whack, miss and round end (once only). Added tests/test_sound.py using a spy object. Passed. Commit: 'Task 4: Add sound feedback'.

Verification

Student: Run everything end to end.

Claude: Headless smoke test of main.py: play at Medium, whack a mole, game over, replay on Hard, game over, Esc to exit - finished cleanly. All three test files pass. Final commit history has one commit per task plus a chore commit that untracks __pycache__.

Final diff summary (starter -> final)

```
 .gitignore                |   2 +
 game/game_engine.py       | 116 ++++++++++++++++++-----
 game/hole.py              |  22 ++---
 game/sounds.py            |  51 ++++++++++
 main.py                   |   2 +
 tests/test_collision.py   |  46 +++++
 tests/test_replay.py      |  59 +++++
 tests/test_sound.py       |  47 +++++
```

Note: this is a structured summary of the session. Export the full raw conversation from the chat interface as well and place it alongside this file if your instructor requires the verbatim transcript.